Web Appendix for Local Networks and Technology Adoption during Industry-Wide Technological Change

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Complete Diagnostic and Specification Results

This Web Appendix reports six tables for completeness. They do not bear on the paper's discussion. Table W.1 reports the preferred network signals for each completed formation block. Table W.2 reports the pooled responses for all four outcomes at every well-to-lease match radius. Table W.3 reports descriptive statistics for the principal signal-specific response samples. Table W.4 reports the sample loss from own lease-expiration restrictions. Table W.5 reports the responses by period for all four outcomes. Table W.6 reports the complete grid of pooled responses with lags one through four. The Internet Appendix retains the evidence that bears on identification, interpretation and robustness.

Table W.1: Preferred formation-block network signals

Formation	Block-closing quarter	Peer performance	Peer skill	Breadth	Technique intensity
Barnett	2007Q1	0.0077 (0.0076)	−0.0194 (0.0178)	0	0.0023 [$F = 1.099$]
Barnett	2008Q1	–	–	1	–
Barnett	2008Q4	0.0207*** (0.0056)	0.0104 (0.0105)	0	–
Barnett	2010Q1	0.0088*** (0.0024)	0.0008 (0.0101)	2	0.0034 [$F = 1.687$]
Barnett	2011Q2	–	–	0	–
Barnett	2012Q4	0.0124 (0.0140)	0.0207 (0.0309)	9	0.0231*** [$F = 3.323$]
Fayetteville	2010Q2	0.0063 (0.0123)	−0.0052 (0.0057)	0	0.0141*** [$F = 5.706$]
Fayetteville	2013Q1	0.0483** (0.0205)	−0.0370 (0.0250)	0	–
Bakken	2011Q3	0.0854*** (0.0140)	−0.0669** (0.0262)	0	–
Bakken	2013Q2	–	–	0	–
Bakken	2014Q4	–	–	1	0.0548*** [$F = 8.776$]
Bakken	2018Q3	−0.0583** (0.0228)	0.3139*** (0.0528)	9	0.0469*** [$F = 6.628$]
Haynesville	2011Q4	0.0482*** (0.0083)	−0.0368*** (0.0130)	0	0.0459*** [$F = 6.220$]
Haynesville	2019Q1	0.0307*** (0.0073)	−0.0338* (0.0177)	3	0.0517*** [$F = 7.266$]
Marcellus	2011Q4	−0.0054 (0.0068)	−0.0431 (0.0462)	0	–
Marcellus	2013Q2	0.0034 (0.0023)	−0.0245 (0.0157)	1	0.0343*** [$F = 4.728$]
Marcellus	2014Q3	0.0212** (0.0091)	−0.0864** (0.0420)	2	0.0342*** [$F = 4.738$]
Marcellus	2016Q1	0.0086 (0.0065)	0.0018 (0.0321)	1	–
Marcellus	2018Q2	0.0234** (0.0111)	−0.0949** (0.0474)	0	0.0295*** [$F = 3.805$]
Marcellus	2020Q2	−0.0136 (0.0118)	0.0355 (0.0506)	0	0.0226*** [$F = 3.254$]
Eagle Ford	2012Q1	0.1172*** (0.0220)	−0.0315* (0.0171)	0	0.0439*** [$F = 5.467$]
Eagle Ford	2012Q4	0.0689*** (0.0135)	−0.0311*** (0.0087)	0	0.0199*** [$F = 2.983$]
Eagle Ford	2013Q2	0.0411*** (0.0098)	−0.0137** (0.0058)	0	0.0152*** [$F = 2.075$]
Eagle Ford	2014Q1	0.0548*** (0.0150)	−0.0206*** (0.0074)	1	0.0846*** [$F = 11.78$]
Eagle Ford	2014Q3	0.0474*** (0.0091)	−0.0242*** (0.0056)	0	0.0579*** [$F = 8.257$]
Eagle Ford	2015Q1	0.0544*** (0.0086)	−0.0522*** (0.0065)	2	0.1048*** [$F = 14.71$]
Eagle Ford	2015Q4	0.0603*** (0.0088)	−0.0320*** (0.0076)	0	0.0414*** [$F = 5.676$]
Eagle Ford	2017Q1	0.0158 (0.0217)	−0.0240* (0.0132)	3	0.0257*** [$F = 3.649$]
Eagle Ford	2018Q2	0.0234 (0.0158)	−0.0168 (0.0199)	9	0.0561*** [$F = 7.816$]

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Table W.1 continued

Formation	Block-closing quarter	Peer performance	Peer skill	Breadth	Technique intensity
Eagle Ford	2019Q2	0.0991*** (0.0155)	−0.1478*** (0.0230)	1	0.0488*** [$F = 6.629$]
Eagle Ford	2020Q4	0.1128*** (0.0251)	−0.2059*** (0.0441)	0	0.0544*** [$F = 7.544$]
Woodford	2012Q3	0.0419*** (0.0162)	0.0146 (0.0319)	0	0.0142** [$F = 1.764$]
Woodford	2017Q2	0.0404*** (0.0056)	0.0170 (0.0148)	4	0.0498*** [$F = 6.991$]
Mississippian	2014Q1	0.0273*** (0.0070)	0.2040*** (0.0292)	11	–
Mississippian	2017Q2	−0.0079 (0.0174)	0.0240 (0.0278)	0	0.0176*** [$F = 2.164$]
Mississippian	2019Q2	0.0059 (0.0123)	−0.0143 (0.0253)	1	0.0274*** [$F = 2.960$]
Three Forks	2014Q2	–	–	1	–
Three Forks	2018Q2	0.0655*** (0.0240)	0.0048 (0.0434)	8	0.0108*** [$F = 2.858$]
Niobrara	2014Q3	0.1144*** (0.0107)	−0.0072 (0.0210)	0	0.0354*** [$F = 4.817$]
Niobrara	2017Q2	0.0732*** (0.0123)	0.0320** (0.0154)	3	0.0502*** [$F = 6.635$]
Niobrara	2019Q2	0.0397*** (0.0090)	0.0180 (0.0167)	7	0.0506*** [$F = 7.185$]
Bone Spring	2015Q1	0.1076*** (0.0240)	−0.1171*** (0.0375)	2	–
Bone Spring	2019Q4	−0.0117 (0.0204)	0.0846 (0.0532)	9	0.0332*** [$F = 4.583$]
Cleveland	2015Q1	0.0025 (0.0185)	0.0120 (0.0299)	15	0.0033*** [$F = 2.130$]
Wolfcamp	2015Q1	0.0570** (0.0258)	−0.0487 (0.0518)	4	0.0246*** [$F = 3.304$]
Wolfcamp	2017Q4	0.0227 (0.0257)	0.0304 (0.0828)	4	0.0557*** [$F = 7.490$]
Wolfcamp	2018Q4	0.0332** (0.0149)	−0.0754* (0.0444)	2	0.0429*** [$F = 6.135$]
Wolfcamp	2019Q4	0.0496*** (0.0099)	−0.1255*** (0.0288)	1	0.0476*** [$F = 6.836$]
Trend Area	2015Q4	0.0925*** (0.0293)	−0.0824* (0.0439)	0	0.0263*** [$F = 3.441$]
Trend Area	2017Q4	0.0408*** (0.0149)	−0.1023*** (0.0312)	1	0.0326*** [$F = 4.867$]
Trend Area	2018Q3	0.0551*** (0.0123)	−0.1173*** (0.0245)	0	0.0272*** [$F = 4.494$]
Trend Area	2019Q3	−0.0056 (0.0118)	0.0042 (0.0235)	1	0.0293*** [$F = 4.306$]
Trend Area	2020Q3	−0.0232** (0.0112)	0.0429* (0.0223)	4	0.0422*** [$F = 5.918$]
Point Pleasant	2020Q4	0.0178* (0.0094)	−0.0261 (0.0365)	1	0.0428*** [$F = 5.843$]

Notes. The unit is a formation by completed 2,000-well block. The block-closing quarter is the final production quarter represented in the block. The resulting signal first enters response observations in the following quarter. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. Peer-performance and peer-skill entries report the spatial coefficient with its standard error in parentheses. Standard errors are the conventional, non-robust standard errors of the generalized spatial two-stage least squares estimator. Breadth is the number of disclosed attributes with positive productivity loadings whose 95 percent intervals exclude zero. Technique-intensity entries report the joint partial R^2 with the Wald F statistic in brackets. Dashes indicate unavailable estimates. Alternating gray bands distinguish formations. $^{*}p < 0.1$, $^{**}p < 0.05$ and $^{***}p < 0.01$.

Table W.2: Pooled responses to expiration-driven peer activity by well-to-lease match radius, all outcomes

Match radius	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Panel A. Peer performance [distance-time]				
0.10 mi.	0.4383***	0.5205***	20.9026	26.87
	(0.1253)	(0.1482)	(13.2349)	(18.71)
0.25 mi. (baseline)	0.1531**	0.1835**	9.4487***	5.62
	(0.0649)	(0.0756)	(2.6951)	(6.70)
0.50 mi.	0.0833**	0.1087**	4.1132***	1.86
	(0.0371)	(0.0512)	(1.4001)	(3.34)
1.00 mi.	0.0510**	0.0661**	1.2204***	1.06
	(0.0242)	(0.0336)	(0.4148)	(2.11)
2.00 mi.	0.0340*	0.0433	1.0225**	1.22
	(0.0206)	(0.0272)	(0.4106)	(1.90)
Full county	0.0299	0.0375	0.8494**	1.14
	(0.0190)	(0.0249)	(0.4221)	(1.84)
Panel B. Peer skill [distance-time]				
0.10 mi.	0.3456***	0.1779***	−31.1305***	−19.83
	(0.0501)	(0.0415)	(5.1246)	(26.38)
0.25 mi. (baseline)	0.0404	0.0272	−1.7864	−37.46**
	(0.1688)	(0.1014)	(2.3706)	(15.57)
0.50 mi.	−0.0069	0.0016	−0.0368	−1.73
	(0.0136)	(0.0207)	(0.0488)	(4.65)
1.00 mi.	0.0003	0.0056	0.1163***	1.86
	(0.0099)	(0.0118)	(0.0385)	(1.53)
2.00 mi.	−0.0013	0.0040	0.1063	1.75
	(0.0102)	(0.0120)	(0.1031)	(1.57)
Full county	−0.0017	0.0035	0.1061	1.75
	(0.0102)	(0.0120)	(0.1030)	(1.57)
Panel C. Peer-technique breadth [equidistant]				
0.10 mi.	−0.0059*	−0.0077**	0.0889	−0.8266***
	(0.0031)	(0.0038)	(0.0829)	(0.1891)
0.25 mi. (baseline)	−0.0020***	−0.0025***	−0.0222	−0.3267*
	(0.0006)	(0.0007)	(0.0367)	(0.1741)
0.50 mi.	−0.0007***	−0.0010***	0.0012	−0.0389
	(0.0003)	(0.0003)	(0.0034)	(0.0340)
1.00 mi.	−0.0005***	−0.0006***	0.0025	−0.0129
	(0.0002)	(0.0002)	(0.0033)	(0.0182)
2.00 mi.	−0.0005***	−0.0006***	0.0020	−0.0104
	(0.0002)	(0.0002)	(0.0035)	(0.0161)
Full county	−0.0005***	−0.0006***	0.0016	−0.0084
	(0.0002)	(0.0002)	(0.0034)	(0.0156)
Panel D. Technique intensity [distance-time]				
0.10 mi.	0.8643***	0.8416**	−9.3348	97.98**
	(0.3225)	(0.3859)	(32.6889)	(44.49)
0.25 mi. (baseline)	0.2661***	0.2988**	8.8524	37.61
	(0.0955)	(0.1249)	(7.3287)	(27.15)
0.50 mi.	0.1054	0.1139	0.6991	8.57
	(0.0777)	(0.0939)	(1.7399)	(6.23)
1.00 mi.	0.0542	0.0593	0.2184	4.36
	(0.0449)	(0.0539)	(0.8414)	(3.15)
2.00 mi.	0.0452	0.0473	0.1644	4.11
	(0.0357)	(0.0428)	(0.6555)	(2.55)
Full county	0.0440	0.0458	0.1779	3.80
	(0.0333)	(0.0396)	(0.6065)	(2.43)

Notes. The unit is a firm–county–formation–quarter. Each panel reports the coefficient on one-quarter-lagged expiration-driven peer activity interacted with the listed network signal. Rows vary the maximum well-to-lease match radius used to construct peer activity. The 0.25-mile radius is the baseline. Standard errors are in parentheses. Every regression includes the lower-order peer-activity and signal terms. Controls include firm and county horizontal-production shares, firm horizontal and total well stocks, the firm's county-level horizontal well stock, a firm-quarter productivity index and annual county size. The regressions absorb operator, county, operator-by-county and calendar-quarter fixed effects. Signal-specific samples require the listed signal to be positive and statistically different from zero. Intensive outcomes require positive activity in the corresponding outcome. Extensive adoption indicates a positive change in the firm's horizontal share. Intensive adoption is the size of that change conditional on positive adoption. Extensive investment indicates positive horizontal drilling. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. Within each signal and outcome the sample size is the same at every radius. The two extensive outcomes use 582,531 observations for peer performance, 152,297 for peer skill, 683,440 for peer-technique breadth and 778,002 for technique intensity. Intensive adoption uses 7,234, 912, 7,058 and 8,704 observations in the same order. Intensive investment uses 8,565, 1,080, 8,420 and 10,348. Standard errors are clustered by formation-block. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table W.3: Descriptive statistics for principal signal-specific response samples

Variable	Mean	SD	*N*
Panel A. Peer-performance response sample			
Extensive adoption	0.0129	0.1127	583,019
Extensive investment	0.0152	0.1223	583,019
Intensive adoption	0.0134	0.9703	583,019
Intensive investment	0.0922	1.3124	583,019
Expiration-driven peer activity (lag 1)	0.0512	0.2307	583,019
Peer performance	0.0461	0.0235	583,019
Panel B. Technique-intensity response sample			
Extensive adoption	0.0115	0.1066	778,269
Extensive investment	0.0137	0.1160	778,269
Intensive adoption	0.0119	0.9112	778,269
Intensive investment	0.0828	1.2024	778,269
Expiration-driven peer activity (lag 1)	0.0358	0.1967	778,269
Technique intensity	0.0324	0.0163	778,269

Notes. The unit is a firm–county–formation–quarter. Panel A restricts the response panel to formation-block observations where peer performance is positive and individually significant. Panel B applies the analogous restriction for technique intensity. These signal-eligible response samples are defined before the complete-case restrictions imposed by each regression. Their observation counts can exceed those reported in the pooled response table. Extensive adoption indicates a positive change in the firm's horizontal share. Intensive adoption is the change in that share conditional on positive adoption. Extensive investment indicates positive horizontal drilling. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. Expiration-driven peer activity is the one-quarter-lagged firm-excluded measure used in the pooled response regressions. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration.

Table W.4: Sample loss from own lease-expiration restrictions

Sample	Rows before	Same-county drop	Any-firm-quarter drop	Same-county drop (%)
Full firm-county-quarter panel	1,005,661	3,142	133,682	0.312
Positive peer activity, 0.25-mile	83,636	559	11,347	0.668
Positive peer-performance signal	585,126	2,107	82,848	0.360
Positive peer-skill signal	153,046	373	22,175	0.244
Positive breadth signal	685,514	1,809	90,503	0.264
Positive intensity signal	780,402	2,133	99,330	0.273
Positive intensive-margin outcome	15,011	601	3,846	4.004

Notes. Rows are samples from the response panel before estimating peer-activity regressions. The same-county restriction drops firm–county–quarters in which the firm has a matched lease expiring in the same county-quarter as the outcome. The any-firm-quarter restriction drops all rows for a firm in a quarter if the firm has any matched lease expiring in that quarter, regardless of county. Positive-signal rows use the preferred signal-specific samples. The positive intensive-margin row is the union of the intensive-adoption and intensive-investment risk sets. The same-county restriction is used in the preferred response sample.

Table W.5: Responses by period across all outcomes

Panel A. Extensive adoption and intensive investment

Signal	Period / contrast	Extensive adoption	Intensive investment
Peer performance	2008–14	−0.0458	10.317
		(0.0573)	(9.463)
	2015–17	0.2822***	11.013
		(0.0780)	(12.761)
	2018+	−0.3274*	66.071***
		(0.1938)	(20.803)
	Rise: (2015–17) − (2008–14)	0.3280***	0.696
		(0.0971)	(16.029)
	Fall: (2018+) − (2015–17)	−0.6096***	55.058**
		(0.1935)	(24.988)
Technique intensity	2008–14	−0.1736***	29.324
		(0.0354)	(22.499)
	2015–17	0.3572*	−18.525
		(0.2107)	(25.286)
	2018+	0.2802*	−24.868
		(0.1447)	(52.654)
	Rise: (2015–17) − (2008–14)	0.5308**	−47.849
		(0.2146)	(32.579)
	Fall: (2018+) − (2015–17)	−0.0770	−6.343
		(0.2451)	(57.570)
Peer-technique breadth	2008–14	−0.0006	0.213
		(0.0005)	(0.159)
	2015–17	−0.0007	−0.397
		(0.0006)	(0.405)
	2018+	−0.0030*	−0.283
		(0.0018)	(0.187)
	Rise: (2015–17) − (2008–14)	−0.0000	−0.610
		(0.0008)	(0.432)
	Fall: (2018+) − (2015–17)	−0.0024	0.114
		(0.0019)	(0.446)
Firm-quarter controls		Yes	Yes
County-quarter controls		Yes	Yes
Era-specific FE and controls		Yes	Yes
Standard errors		formation-block	formation-block

Table W.5 (continued): Responses by period across all outcomes

Panel B. Extensive investment and intensive adoption

Signal	Period / contrast	Extensive investment	Intensive adoption
Peer performance	2008–14	0.0009	1.715
		(0.0702)	(1.776)
	2015–17	0.2807***	15.422
		(0.0919)	(9.726)
	2018+	−0.3430	−0.236
		(0.2608)	(3.415)
	Rise: (2015–17) − (2008–14)	0.2798**	13.707
		(0.1167)	(9.241)
	Fall: (2018+) − (2015–17)	−0.6237**	−15.658
		(0.2521)	(10.396)
Technique intensity	2008–14	−0.1530**	4.782
		(0.0740)	(3.779)
	2015–17	0.2962	7.995
		(0.2427)	(15.994)
	2018+	0.3191**	4.019
		(0.1599)	(27.112)
	Rise: (2015–17) − (2008–14)	0.4491*	3.213
		(0.2512)	(16.693)
	Fall: (2018+) − (2015–17)	0.0229	−3.976
		(0.2773)	(31.621)
Peer-technique breadth	2008–14	−0.0009*	−0.035
		(0.0005)	(0.027)
	2015–17	−0.0007	−0.238
		(0.0007)	(0.159)
	2018+	−0.0039**	0.031
		(0.0019)	(0.033)
	Rise: (2015–17) − (2008–14)	0.0002	−0.203
		(0.0010)	(0.162)
	Fall: (2018+) − (2015–17)	−0.0031	0.269
		(0.0020)	(0.165)
Firm-quarter controls		Yes	Yes
County-quarter controls		Yes	Yes
Era-specific FE and controls		Yes	Yes
Standard errors		formation-block	formation-block

Notes. Panel A reports extensive adoption and intensive investment. Panel B reports extensive investment and intensive adoption. The interaction coefficient is estimated at the one-quarter lag, separately by period, under the separate-fit specification of the main paper. Every regression includes the lower-order peer-activity and signal terms, the main controls and operator, county, operator-by-county and calendar-quarter fixed effects. The sample excludes firm-county-quarters in which the responding firm has a same-county lease expiration. The rise and fall rows are directly tested contrasts. Standard errors are clustered by formation-block. Each specification uses the formation-block observations in which its own signal is positive and individually significant. Extensive adoption indicates a positive change in the firm's horizontal share. Intensive adoption is the size of that change conditional on positive adoption. Extensive investment indicates positive horizontal drilling. Intensive investment is the number of horizontal wells conditional on positive horizontal investment. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table W.6: Pooled peer-activity responses with lags one through four, complete grid

	Extensive adoption	Extensive investment	Intensive adoption	Intensive investment
Peer performance [distance-time]				
$L1$	0.1103^{**}	0.1279^{***}	6.5339^{***}	10.8842^{**}
	(0.0439)	(0.0459)	(1.5849)	(4.5347)
$L2$	0.0332	0.0852	−1.0503	6.4468
	(0.0755)	(0.0814)	(3.1858)	(8.6082)
$L3$	0.1392^{***}	0.1549^{***}	15.3146^{*}	−17.6677
	(0.0392)	(0.0505)	(9.0080)	(14.3150)
$L4$	0.0905	0.0650	7.0446	−32.1251
	(0.0675)	(0.0803)	(4.5236)	(27.2803)
Peer skill [distance-time]				
$L1$	−0.0265	−0.0615	−0.8432	-40.2744^{*}
	(0.1648)	(0.0956)	(1.9988)	(14.4138)
$L2$	0.0071	−0.1023	0.3921	−21.6694
	(0.2221)	(0.0925)	(2.8589)	(28.6237)
$L3$	−0.0107	−0.0321	1.2493	−3.0252
	(0.0853)	(0.1793)	(0.9409)	(10.1111)
$L4$	-0.2179^{*}	−0.3123	2.7351^{**}	−1.1663
	(0.0775)	(0.1968)	(0.7408)	(13.8659)
Peer-technique breadth [equidistant]				
$L1$	-0.0008^{**}	-0.0012^{***}	−0.0218	−0.1352
	(0.0004)	(0.0004)	(0.0352)	(0.2137)
$L2$	-0.0009^{***}	-0.0009^{***}	−0.0208	0.0108
	(0.0003)	(0.0003)	(0.0312)	(0.1223)
$L3$	-0.0007^{***}	-0.0008^{***}	0.0185	0.0111
	(0.0002)	(0.0002)	(0.0168)	(0.0910)
$L4$	-0.0011^{**}	-0.0012^{***}	-0.0700^{***}	−0.1261
	(0.0004)	(0.0004)	(0.0225)	(0.0932)
Technique intensity [distance-time]				
$L1$	0.2477^{***}	0.2882^{***}	6.1222	40.8280
	(0.0884)	(0.1052)	(6.7062)	(28.6120)
$L2$	0.0257	0.0135	6.8601^{*}	3.7816
	(0.0712)	(0.0878)	(3.7108)	(11.7856)
$L3$	0.0663	0.0581	0.5296	1.4570
	(0.0883)	(0.0974)	(5.2048)	(15.3864)
$L4$	−0.0107	−0.0417	4.7289	−28.9124
	(0.0900)	(0.0997)	(4.2488)	(24.3881)

Notes. The unit is a firm–county–formation–quarter. Rows report the coefficient on expiration-driven peer activity at lags one through four interacted with the listed signal, with all four lags entered jointly. L1 through L4 denote peer activity lagged one through four quarters. Each regression includes the corresponding lower-order peer-activity terms, the signal term, baseline controls and fixed effects, the signal-specific sample and the same-county own-expiration restriction. Intensive outcomes condition on positive activity in the corresponding outcome. Peer performance, peer skill and technique intensity use distance-time weights. Peer-technique breadth uses equidistant weights. Standard errors cluster by formation-block. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.